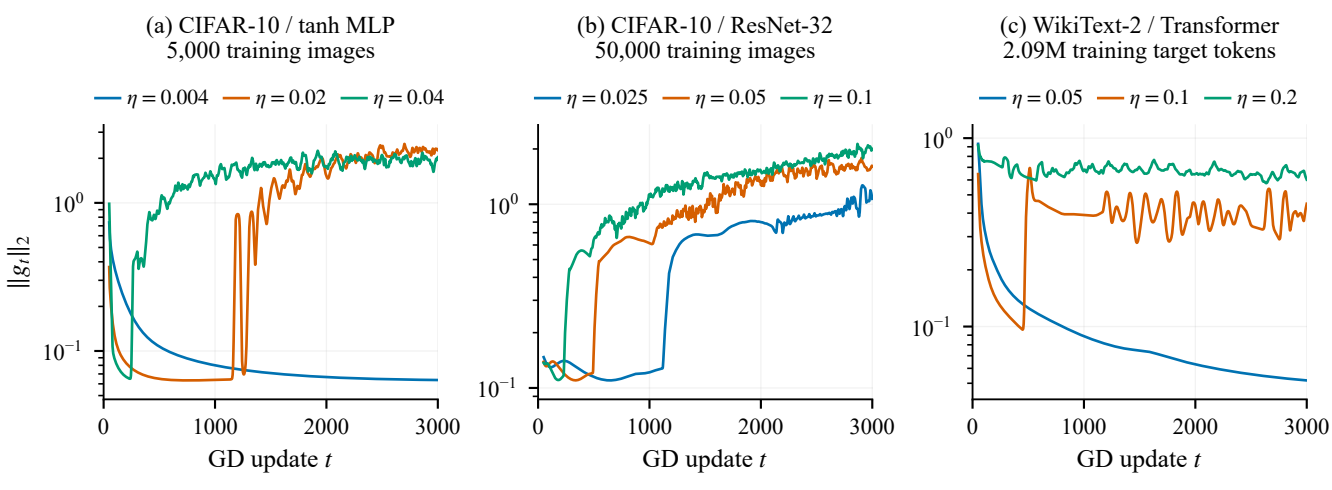

(a) CIFAR-10 / tanh MLP
5,000 training images
(b) CIFAR-10 / ResNet-32
50,000 training images
(c) WikiText-2 / Transformer
2.09M training target tokens
$\eta = 0.004$
$\eta = 0.02$
$\eta = 0.04$
$\eta = 0.025$
$\eta = 0.05$
$\eta = 0.1$
$\eta = 0.05$
$\eta = 0.1$
$\eta = 0.2$
$\|g_t\|_2$
10^0
10^{-1}
0
1000
2000
3000
GD update t